CSR Analysis – Weekly Passenger Flows & Energy Estimates

1. Weekly Passenger Flow Overview

- **Low weeks (Green – Not busy):**
  - Typical early-year weeks (January–February) and late-year (November–December).
  - Passenger numbers: **~130k–170k per week**, **energy ~65–85k kWh.**
- **Moderate weeks (Yellow – Busy):**
  - Spring and fall transition weeks (March–May, December).
  - Passenger numbers: **~180k–200k per week**, **energy ~90–100k kWh.**
- **Peak weeks (Red – Very busy):**
  - Summer and key holiday weeks.
  - Passenger numbers: **~200k–230k per week**, **energy ~100–115k kWh.**

2. Patterns & Insights

- **Seasonality is clear:**
  - **Winter weeks are consistently green (off-peak).**
  - **Summer weeks (June–August)** show red peaks.
  - **Spring (April–May) and autumn (October)** are mostly Yellow with some Red spikes.
- **Energy correlation:**
  - Weekly energy consumption scales directly with passenger numbers:
    - Green ~50% less than Red weeks.
    - Yellow ~10–15% below Red weeks.
- **Special events/holidays:**
  - Peaks around **late April, July, and December–January** indicate high holiday traffic.

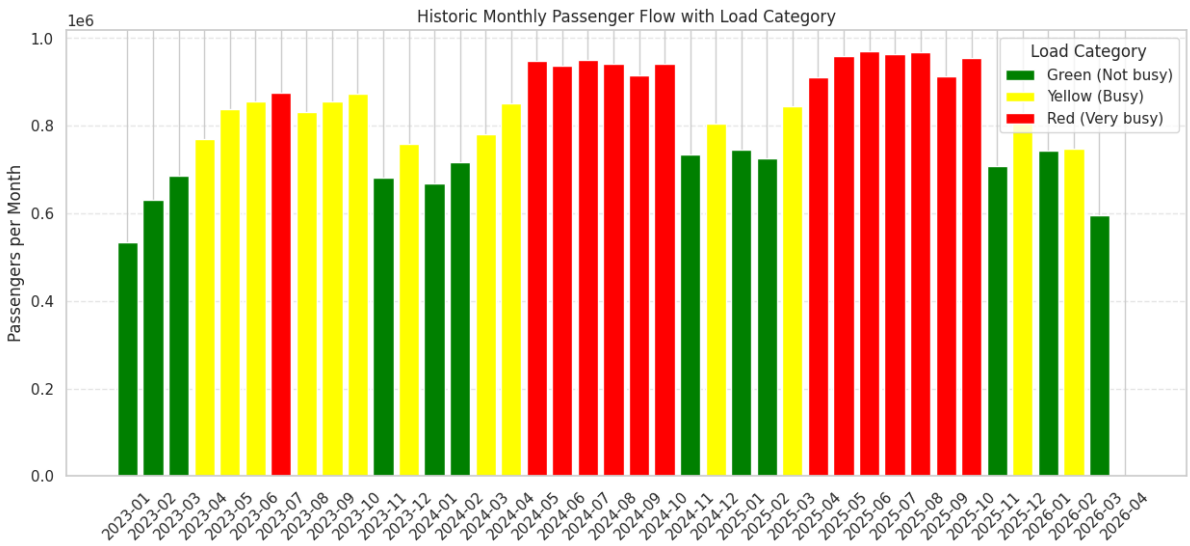
3. Operational Recommendations

1. **Staffing & resources:**
   - Red weeks require full staffing and operational readiness.
   - Yellow weeks: moderate resources.
   - Green weeks: minimal staffing; ideal for maintenance and optimization.
2. **Energy efficiency:**
   - Consider dynamically scaling lighting, HVAC, and equipment based on weekly passenger estimates.

4. Yearly Overview – 2026 Monthly Status

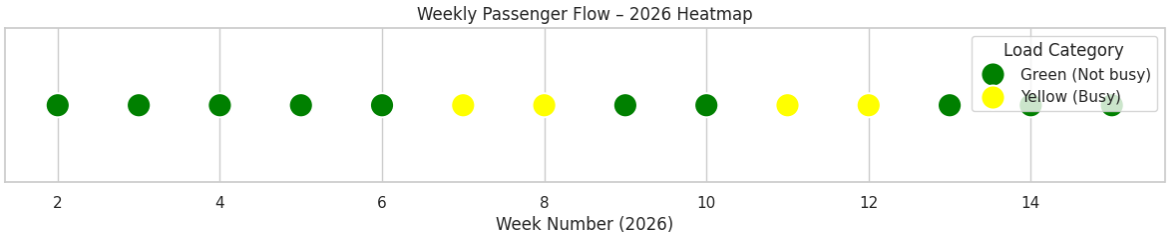
Month	Status (Predominant)
January	Less Busy (Green)
February	Less Busy / Busy (Mixed)

Month	Status (Predominant)
March	Less Busy / Busy (Mixed)
April	Busy / Very Busy (Yellow & Red)
May	Very Busy (Red dominates)
June	Very Busy (Red dominates)
July	Very Busy (Red dominates)
August	Very Busy / Busy (Mostly Red, some Yellow)
September	Very Busy / Busy (Red early, Yellow later)
October	Busy (Yellow dominates)
November	Less Busy (Green dominates)
December	Busy (Yellow dominates, approaching holiday peaks)



5. Predictions for 2026 Based on Historical Data

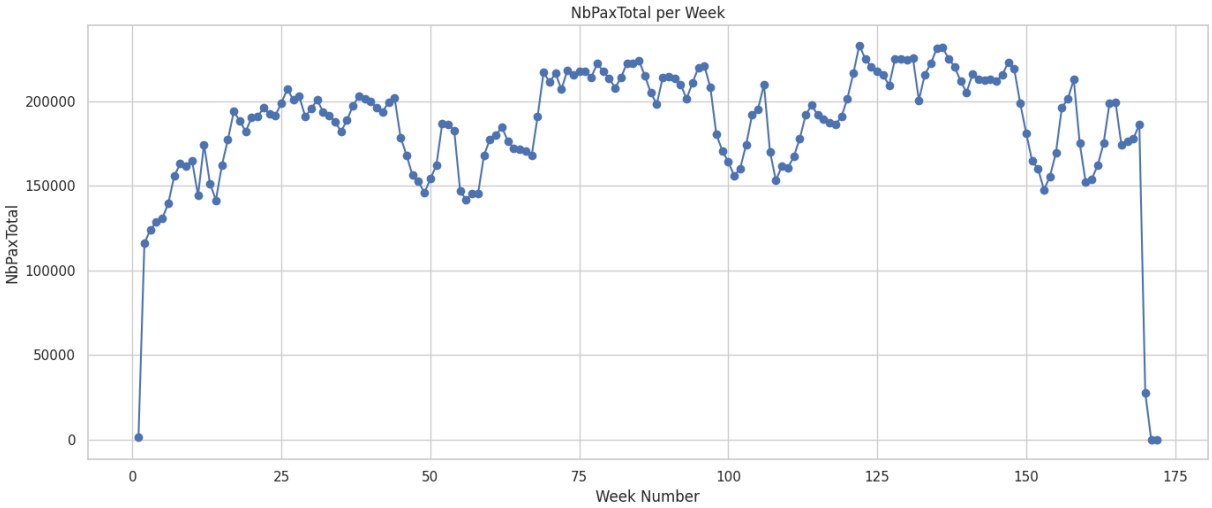
- **January–February:** Low traffic, mostly Green weeks → minimal staffing needed.
- **March–April:** Transition from Green to Yellow → moderate staffing; anticipate energy usage rising.
- **May–August:** Peak months; Red weeks dominate → full operational readiness and peak energy consumption expected.

- 

- **September–October:** Slight decrease, Yellow weeks dominate → plan for moderate operations.
- **November–December:** Traffic rises in December (holiday season) → anticipate busy weeks; late November quieter.

Insights for planning:

- Energy demand is expected to **mirror passenger flows**.
- Red weeks require advanced coordination of staff, energy systems, and maintenance schedules.
- Historical trends suggest **late April, summer holidays, and end-of-year weeks** will consistently be Very Busy.



Variation of number of passengers per week over the last 175 weeks